

## 1 Highlights

### 2 **Sentinel-2 and Landsat 8/9 retrieval of water transparency in Lake** 3 **Titicaca: harmonization limits, validation design and calibrated** 4 **uncertainty**

5 Dina Maribel Yana-Yucra, Richar Andre Vilca-Solorzano, Fred Torres-Cruz,  
6 Vladimiro Ibañez

- 7     • Land-derived Landsat/Sentinel-2 harmonization is invalid over dark  
8       oligotrophic water
- 9     • Sensors stay 99.8% separable after harmonization; intercepts exceed  
10       signal 70-fold
- 11    • Campaign, not station, is the blocking unit in campaign-based moni-  
12       toring programmes
- 13    • Honest Secchi retrieval:  $R^2=0.574$ , RMSE 1.86 m over 812 real match-  
14       ups
- 15    • Conformal intervals are calibrated marginally but under-cover in the  
16       clearest water

17        Sentinel-2 and Landsat 8/9 retrieval of water  
18        transparency in Lake Titicaca: harmonization limits,  
19        validation design and calibrated uncertainty

20        Dina Maribel Yana-Yucra<sup>a</sup>, Richar Andre Vilca-Solorzano<sup>a,\*</sup>, Fred  
21        Torres-Cruz<sup>a</sup>, Vladimiro Ibañez<sup>a</sup>

*<sup>a</sup>Faculty of Statistical and Computer Engineering, Universidad Nacional del Altiplano  
(UNAP), Av. Floral 1153, Puno, 21001, Peru*

---

22        **Abstract**

23        Monitoring water transparency in high-altitude oligotrophic lakes is critical  
24        for freshwater management, yet it remains constrained by the sparsity of field  
25        data. While multi-mission satellite archives (e.g., harmonized Sentinel-2 and  
26        Landsat 8/9) and machine learning offer a synoptic solution, current practices  
27        often import assumptions from turbid-water literature that fail in clear-water,  
28        campaign-based settings. We assemble 1002 satellite–field match-ups against  
29        twelve years of limnological monitoring in Lake Titicaca to address these fail-  
30        ures and define a defensible retrieval protocol. First, we demonstrate that  
31        standard land-derived harmonization coefficients (e.g., Roy et al., 2016) are  
32        invalid over dark water, as their intercepts exceed the native Landsat signal  
33        by factors of 1.4 to 69.8, leaving sensors 99.8% separable; we propose a clas-  
34        sifier two-sample test to catch this failure. Second, we show that standard  
35        station-blocked cross-validation fails to prevent spatial leakage in campaign-  
36        based sampling: under campaign-blocked folds, a station-mean null model  
37        still explains  $R^2 = 0.208$  of the variance. We introduce a null-model diag-

---

\*Corresponding author: Richar Andre Vilca-Solorzano.

*Email addresses:* 75923467@est.unap.edu.pe (Dina Maribel Yana-Yucra),  
75521963@est.unap.edu.pe (Richar Andre Vilca-Solorzano)

nostic to identify the correct blocking unit (the campaign). Finally, under this strict design, we train a Random Forest regressor to retrieve Secchi disk depth ( $R^2 = 0.574$ , RMSE 1.86 m) and attach split-conformal prediction intervals. We reveal that while marginally calibrated (89.4% coverage at 90% nominal), conformal intervals structurally under-cover in the clearest water (71.3%). Our findings provide a rigorous blueprint for evaluating machine-learning water quality products in large, oligotrophic lakes.

*Keywords:* Lake Titicaca, Secchi disk depth, Sentinel-2, Landsat, cross-sensor harmonization, conformal prediction

---

## 1. Introduction

Large lakes are sentinels of environmental change, but continuous monitoring of remote, high-altitude systems is constrained by sparse field infrastructure (Williamson et al., 2009). Lake Titicaca (3,810 m a.s.l.), the largest tropical lake in the world by volume, is a critical transboundary freshwater reserve whose water quality is documented only by infrequent field campaigns (OEFA, 2016; Siguayro Mamani et al., 2022), even as eutrophication advances in its near-shore bays (Heredia et al., 2022; Lanza et al., 2024). Satellite remote sensing offers a synoptic solution (Gholizadeh et al., 2016; Topp et al., 2020). Specifically, water transparency—quantified by the Secchi disk depth ( $Z_{SD}$ )—is an integrative indicator of water clarity strongly linked to light attenuation (Lee et al., 2015). Because Sentinel-2 MSI and Landsat 8/9 OLI provide complementary revisit times, their joint use as a virtual constellation coupled with machine learning is now the standard practice for mapping water quality (Pahlevan et al., 2020; Cao et al., 2022).

62 Although these virtual constellations work well in turbid, productive  
 63 “Case-2” waters, they fail to transfer reliably to clear, oligotrophic high-  
 64 altitude lakes due to two unresolved methodological assumptions. First,  
 65 the cross-sensor harmonization coefficients required to merge Landsat and  
 66 Sentinel-2 (e.g., Roy et al., 2016) were fitted on terrestrial targets. Because  
 67 terrestrial surfaces are orders of magnitude brighter than clear water, their  
 68 additive intercepts overwhelm the over-water signal. While near-infrared  
 69 bias over water has been noted (Pahlevan et al., 2019), the magnitude of  
 70 this mismatch and its effect on machine-learning retrievals remain unquan-  
 71 tified. Second, to prevent spatial autocorrelation from inflating apparent  
 72 accuracy, previous methods routinely block cross-validation by monitoring  
 73 station (Roberts et al., 2017). However, they assume rather than test that  
 74 station identity is the primary source of data leakage. In campaign-based  
 75 monitoring programmes, where an entire lake is sampled within a few days  
 76 under identical conditions, station-blocking may retain severe leakage and  
 77 falsely inflate performance metrics.

78 In this paper, we propose a rigorous validation and retrieval pipeline to  
 79 overcome these challenges, using Lake Titicaca as our testing ground. Our  
 80 innovation is replacing assumed methodological defaults with data-driven  
 81 diagnostics. Specifically, we assemble 1002 real satellite–field match-ups and  
 82 apply a classifier two-sample test (adversarial validation) (Lopez-Paz and  
 83 Oquab, 2017) to diagnose harmonization failure. Furthermore, we introduce  
 84 a null-model diagnostic to identify the true blocking unit for cross-validation.  
 85 Finally, we train a Random Forest regressor under the corrected design and  
 86 attach split-conformal prediction intervals (Angelopoulos and Bates, 2023)

87 to quantify per-prediction uncertainty.

88 In contrast to previous water-quality mapping efforts (Ayala Izurieta  
89 et al., 2023; Maligaya et al., 2024), our study makes three explicit contri-  
90 butions:

- 91 **1. Quantification of Harmonization Limits:** We show that land-  
92 derived harmonization intercepts exceed the native Landsat signal by  
93 up to 69.8-fold over clear water. Consequently, a classifier can identify  
94 the source sensor with 99.8% accuracy from harmonized bands, proving  
95 that the model learns a sensor label rather than unified optics.
- 96 **2. Null-Model Validation Design:** We demonstrate that under campaign-  
97 blocked folds, a station-mean null model still achieves  $R^2 = 0.208$ ,  
98 proving that station-blocking fails to prevent leakage. We establish the  
99 campaign as the required blocking unit for this class of data.
- 100 **3. Defensible Retrieval with Conformal Uncertainty:** Under this  
101 strict design, our model achieves an honest  $R^2$  of 0.574 (RMSE 1.86 m).  
102 We further reveal that while split-conformal intervals are marginally  
103 calibrated (89.4% coverage), they structurally under-cover extreme val-  
104 ues in the clearest water, highlighting a critical limitation for opera-  
105 tional monitoring.

## 106 2. Materials and methods

### 107 2.1. Study area

108 Lake Titicaca comprises three limnologically distinct zones (Fig. 1): the  
109 large, deep, oligotrophic Lago Mayor; the shallower Lago Menor (Wiñay-  
110 marca) to the south-east; and Bahía de Puno, which receives municipal and

111 agricultural inputs from the city of Puno and is the most eutrophic sector.  
112 Measured transparency spans 1.5 m in near-shore turbid water to 16.5 m in  
113 the open oligotrophic basin.

## 114 2.2. *In-situ data*

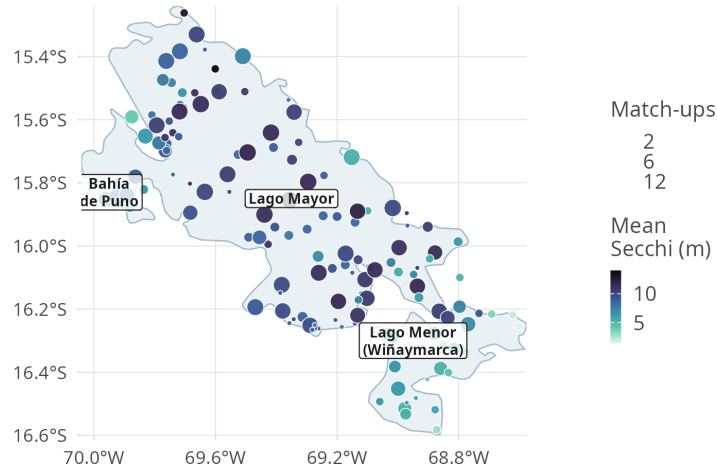
115 We used 881 field measurements of Secchi transparency, with co-measured  
116 chlorophyll-*a*, suspended solids, temperature, pH, dissolved oxygen and nu-  
117 trients, from the limnological monitoring campaigns of IMARPE (Instituto  
118 del Mar del Perú) on Lake Titicaca, spanning 2011–2024 across 160 sta-  
119 tion identifiers distributed over the whole lake. Six of those identifiers are  
120 a 2017 naming variant of a station already recorded under another label,  
121 so the records describe 154 distinct stations; we retain the identifiers as  
122 issued, which makes station-blocked validation marginally more permissive  
123 and therefore does not favour the argument of Section 2.6. Of the 881 mea-  
124 surements, 734 include a valid Secchi reading. The July 2019 campaign (71  
125 stations) is published by Siguyayro Mamani and Franco (2022); the remaining  
126 years derive from the same IMARPE programme. As a provenance check, our  
127 2019 zone-level means reproduce the published values almost exactly (Lago  
128 Mayor conductivity 1513.4 vs. 1513.5  $\mu\text{S cm}^{-1}$ ; Lago Menor chlorophyll-*a*  
129 1.52 vs. 1.52  $\text{mg m}^{-3}$ ).

130 Three properties of this record constrain everything that follows, and we  
131 state them here rather than in the limitations (Fig. A.5).

132 First, the monitoring programme is itself almost entirely a dry-season  
133 programme. The 2011–2024 record contains 13 campaigns with Secchi read-  
134 ings, of which exactly one — December 2012, 72 measurements — falls in the  
135 austral wet season (December–March). That campaign predates both sen-

**(a) Limnological monitoring network (IMARPE)**

IMARPE/ALT stations with satellite-field match-ups (2013–2024), coloured by mean measured Secchi depth. The trophic gradient is visible in the field data alone



**(b) Sampling effort is very uneven**

Each point is a station. 143 stations, but most were visited in very few campaigns, limiting the power of any station-based analysis

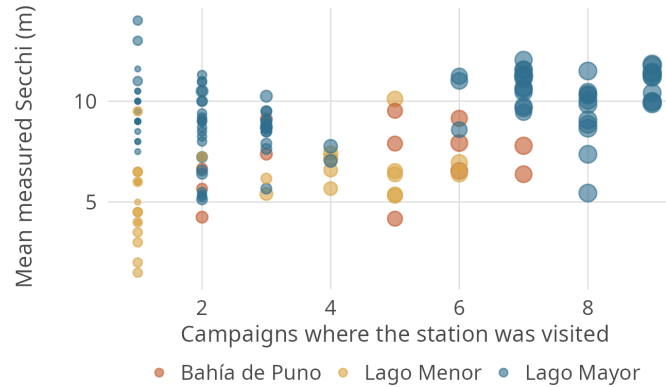


Figure 1: Study area and monitoring network. (a) IMARPE/ALT stations with a satellite–field match-up, coloured by mean measured Secchi depth over the Natural Earth lake boundary; the trophic gradient is evident in the field data alone. (b) Sampling effort per station: most stations were visited in few campaigns, which limits the power of any station-level analysis.

sors used here: Landsat 8 became operational in May 2013 and Sentinel-2A in June 2015. Consequently every match-up in our dataset falls between July and October, and this is a property of the field programme rather than of our processing: no choice of match-up window or cloud threshold can recover wet-season pairs that have no contemporaneous imagery. We return to what this means for operational use in Section 4.3.

Second, sampling is episodic: campaigns exist for 2013–2024 with no campaigns in 2020, 2021, 2023.

Third, the reference measurement is coarse: 73 distinct Secchi values occur in 812 match-ups, because disk readings are recorded to the nearest half metre.

No synthetic or spectrally derived labels were used at any stage.

### *2.3. Satellite data and cross-sensor harmonization*

Sentinel-2 MSI (harmonized surface reflectance) and Landsat 8/9 OLI (Collection 2, Level-2; Vermote et al., 2016) were accessed through Google Earth Engine (Gorelick et al., 2017). Clouds were masked with the Sentinel-2 Scene Classification Layer plus cloud probability, and with the Landsat QA\_PIXEL bitmask (dilated cloud, cloud and shadow). The cirrus flag was not applied, as it is near-permanently set at 3810 m and would mask the entire archive. We retained the six bands common to both sensors (blue, green, red, NIR, SWIR1, SWIR2).

Landsat surface reflectance was first harmonized to Sentinel-2 equivalent reflectance with the per-band linear coefficients of Roy et al. (2016), of the form  $\rho_{\text{MSI}} = a + b \rho_{\text{OLI}}$ . Section 3.1 evaluates whether this transform is valid over the water of Lake Titicaca and derives an alternative. For that evalu-

161 ation we identified every in-situ event observed by both sensors within the  
162 match-up window — 270 paired station–date observations — and regressed  
163 Sentinel-2 on harmonized Landsat reflectance band by band.

#### 164 *2.4. Satellite–field match-ups*

165 For every in-situ measurement we extracted the cloud-masked median  
166 reflectance within a 30 m buffer of the station coordinate, from imagery  
167 acquired within  $\pm 10$  days. The 30 m buffer matches the coarser Landsat pixel  
168 and averages residual geolocation error and sensor noise over several pixels  
169 while remaining in open water, away from shoreline mixing. The  $\pm 10$ -day  
170 window balances the scarce cloud-free image availability at 3810 m against  
171 the slow temporal change of transparency in this large, deep lake, and is  
172 consistent with windows used in operational  $Z_{SD}$  retrieval (Kloiber et al.,  
173 2002; Zhang et al., 2022).

174 This yielded 1002 match-ups (340 Sentinel-2, 662 Landsat), of which 812  
175 have a Secchi value, across 143 stations and 120 campaign dates. Because  
176 both sensors can observe the same field event, 272 match-ups are cross-sensor  
177 pairs of a single in-situ measurement; collapsing them leaves 540 independent  
178 events, and Section 3.2 verifies that accuracy is unchanged, confirming the  
179 result is not an artefact of pseudo-replication.

#### 180 *2.5. Features and model*

181 Twelve sensor-agnostic features were computed from the six common  
182 bands: the bands themselves plus NDWI, NDTI and the ratios blue/green,  
183 green/blue, red/green and blue/red. We trained a Random Forest regres-  
184 sor (Breiman, 2001). Reflectance was scaled with a robust (median/IQR)

185 scaler fitted only on the training partition of each fold. All models were  
186 implemented with scikit-learn (Pedregosa et al., 2011).

187 Model choice was settled empirically rather than assumed. We com-  
188 pared Random Forest, Extremely Randomized Trees, XGBoost (Chen and  
189 Guestrin, 2016) and LightGBM, each with hyperparameters optimized by  
190 Optuna (Akiba et al., 2019) inside *nested* cross-validation — tuning on an  
191 inner campaign-blocked split of each outer training partition, so no outer fold  
192 ever informs its own hyperparameters — and benchmarked the resulting gain  
193 against the spread across random seeds. Section 3.4 reports the outcome.

## 194 2.6. Validation design and null models

195 Rather than a single split, we evaluate a hierarchy of increasingly de-  
196 manding designs (Roberts et al., 2017): random  $K$ -fold; GroupKFold blocked  
197 by station; GroupKFold blocked by *campaign date*; GroupKFold blocked by  
198 year; a temporal hold-out; and leave-one-zone-out, in which an entire trophic  
199 zone is withheld.

200 Crucially, we do not assume which blocking unit matters. We benchmark  
201 each design against null models that predict, respectively, the training-set  
202 mean of the zone, of the station and of the campaign date. A blocking unit  
203 is doing real work if, and only if, the corresponding null model collapses  
204 under it. This diagnostic, rather than convention, determines our primary  
205 design.

206 Performance is reported as out-of-fold  $R^2$ , RMSE and MAE, and bench-  
207 marked against a classical log-linear blue/green band-ratio algorithm in the  
208 standard form  $\ln Z_{SD} \sim \ln(\rho_{\text{blue}}/\rho_{\text{green}})$  (Kloiber et al., 2002), with predic-  
209 tions bounded to the physically observable range, and against a multiband

210 linear model.

## 211 2.7. Uncertainty and interpretation

212 We attach a per-prediction interval using split-conformal prediction (An-  
213 gelopoulos and Bates, 2023): within each fold, a calibration subset of the  
214 training partition yields the  $(1 - \alpha)$  quantile of absolute residuals, defining  
215 a symmetric interval with finite-sample marginal coverage guarantees (Ro-  
216 mano et al., 2019). Because a single nominal level can be met by chance, we  
217 report empirical coverage at 50, 80, 90 and 95%, and also coverage condi-  
218 tional on measured transparency, since marginal calibration does not imply  
219 conditional calibration.

220 The model is interpreted with SHAP (Lundberg and Lee, 2017; Lundberg  
221 et al., 2020), computed *out of fold* — each observation explained by the model  
222 that did not see it — rather than on a model fitted to the whole dataset, which  
223 inflates the apparent importance of memorized features. Trends in annual  
224 median in-situ transparency were tested with Mann–Kendall and Sen’s slope  
225 (Mann, 1945; Sen, 1968) on the full 2011–2024 in-situ record, not on the  
226 match-up subset.

## 227 3. Results

### 228 3.1. Land-derived harmonization is invalid over oligotrophic water

229 The Roy et al. (2016) transforms carry additive intercepts of 0.0116–  
230 0.0448 in reflectance units. Over Lake Titicaca the native Landsat over-water  
231 signal is far smaller than this (Table A.2, Fig. 2a). In the near-infrared the  
232 median native reflectance is effectively zero, so the intercept exceeds the

233 signal by a factor of 69.8; in SWIR1 the factor is 23.2, in red 10.6, and even  
 234 in blue — the brightest water band — 2.5. The harmonized Landsat bands  
 235 are consequently dominated by a constant inherited from a land calibration:  
 236 their medians exceed the corresponding Sentinel-2 medians by 1.6-fold in  
 237 green and 15.4-fold in the near-infrared.

238 A harmonization that succeeded would make the two sensors statistically  
 239 indistinguishable over the same water. They are not. A Random Forest clas-  
 240 sifier trained to identify the source sensor from the harmonized features alone  
 241 reaches 99.8% accuracy against a 66.3% majority-class baseline (Fig. 2b); re-  
 242 stricting it to the visible bands gives 97.9%, and to scale-invariant band ratios  
 243 99.5%. This is a classifier two-sample test (Friedman, 2003; Lopez-Paz and  
 244 Oquab, 2017): if two samples come from the same distribution, no classifier  
 245 can separate them above chance. Sensor identity therefore remains fully en-  
 246 coded in the harmonized reflectance, and any model trained on the pooled  
 247 archive can read it. The test is not new, but to our knowledge it has not  
 248 been used as a harmonization acceptance check; it costs one model fit and it  
 249 is decisive.

250 Refitting locally on the 270 station–date events observed by both sensors  
 251 gives water-specific coefficients (Table A.3). These restore the spectral sig-  
 252 nature almost exactly — the median Landsat/Sentinel-2 ratio falls from 15.4  
 253 to about 1.0 across all six bands (Fig. 2d) — but per-observation agreement  
 254 remains uneven (Fig. 2c). Green recovers usable agreement ( $r = 0.71$ ), and  
 255 NIR and SWIR reach  $r \approx 0.6$ . Blue does not:  $r = 0.26$ . This is the most  
 256 consequential result of the section, because SHAP attributes 72.6% of the  
 257 retrieval to green/blue bands and ratios (Section 3.7). The band on which

258 the retrieval most depends is the band on which the two sensors least agree.

### 259 3.2. *The campaign, not the station, is the blocking unit*

260 Table 1 reports the retrieval under each validation design alongside the  
261 null models (Fig. A.6). Blocking by station changes nothing relative to a ran-  
262 dom split (both  $R^2 = 0.596$ ), which under the usual reading would be taken  
263 as evidence of no spatial leakage. The null models show why that reading  
264 is wrong: under campaign-blocked folds the station-mean null still attains  
265  $R^2 = 0.208$  and the zone-mean null  $R^2 = 0.287$ , whereas the campaign-date-  
266 mean null collapses to  $-0.012$ . Dependence in this dataset lives between  
267 campaigns, not between stations, because a single campaign samples the  
268 whole lake within a few days under one set of atmospheric and limnological  
269 conditions. Station-blocked folds leave that structure intact.

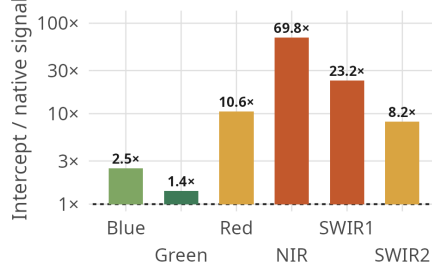
270 Under campaign-blocked validation — our primary design — the Random  
271 Forest attains  $R^2 = 0.574$  and RMSE 1.86 m over 812 match-ups, essentially  
272 unbiased ( $-0.02$  m). Blocking by year is stricter still ( $R^2 = 0.495$ ). Collaps-  
273 ing cross-sensor pseudo-replicates to 540 independent events leaves accuracy  
274 unchanged ( $R^2 = 0.571$ ).

### 275 3.3. *Retrieval accuracy and benchmarks*

276 Against this design the classical blue/green band ratio attains  $R^2 = 0.097$   
277 (RMSE 2.71 m) and a multiband linear model  $R^2 = 0.323$  (RMSE 2.34 m)  
278 (Table A.4, Fig. 3). We report one methodological caution alongside these  
279 figures. An unbounded implementation of the same band-ratio algorithm —  
280 log-transformed and back-transformed without constraining predictions to  
281 the observable range — produces a single prediction of 143.5 m against an

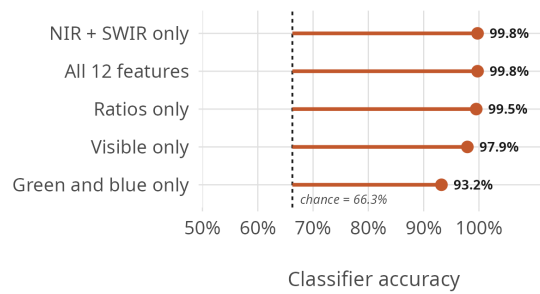
**(a) Land intercept exceeds the signal**

Roy et al. (2016) coefficients, fitted over vegetation and soil, vs native Landsat reflectance over the lake water. The dashed line marks intercept = signal



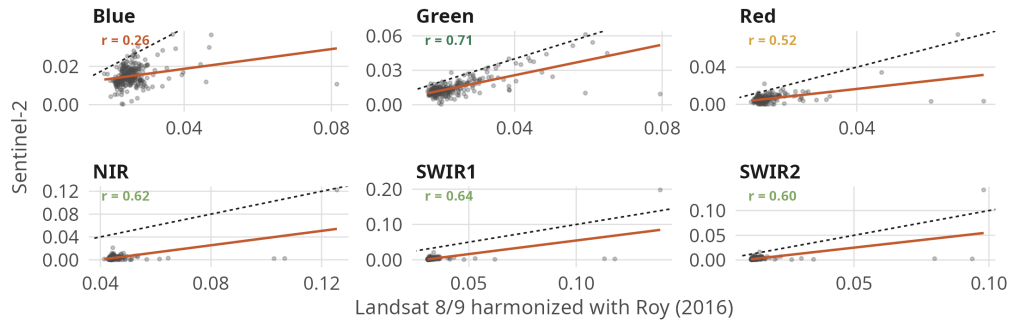
**(b) Sensors remain distinguishable**

Accuracy of a Random Forest predicting the source sensor from already harmonized reflectance. If harmonization worked, it would be at chance



**(c) Blue is the band that agrees least between sensors**

Sentinel-2 reflectance vs Roy-harmonized Landsat in the 270 events observed by both sensors at the same station and date. The dashed line is 1:1



**(d) Local recalibration over water does harmonize bands**

Median reflectance over water by sensor. Labels are the Landsat / Sentinel-2 ratio: it goes from 15.4x in NIR to ~1.0x in every band

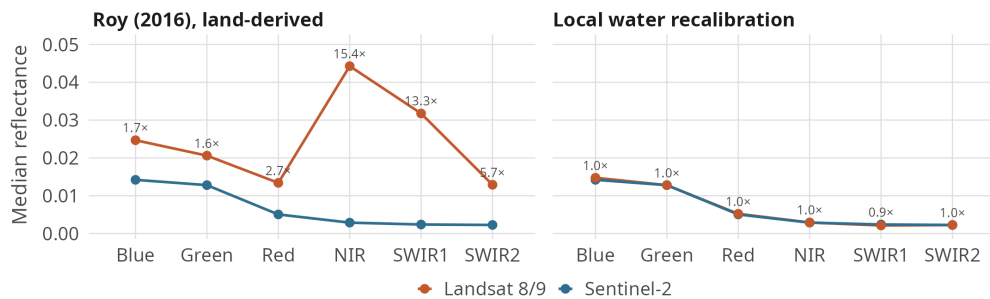


Figure 2: Why land-derived harmonization fails over clear water. (a) The Roy intercept relative to the native over-water Landsat signal, by band. (b) Accuracy of a classifier predicting the source sensor from harmonized features; under a successful harmonization it would sit at chance. (c) Sentinel-2 versus harmonized Landsat reflectance on the 270 paired station-date events; blue agrees worst. (d) Median spectral signature before and after local water recalibration; labels are the Landsat/Sentinel-2 ratio.

Table 1: Validation design, null models and blocking units (Random Forest,  $n = 812$  unless noted). A blocking unit is effective only if the corresponding null model collapses under it.

|                    | Setting                                            | $R^2$        | RMSE (m)    |
|--------------------|----------------------------------------------------|--------------|-------------|
| Null models        | Zone-mean                                          | 0.287        | 2.40        |
|                    | Station-mean                                       | 0.208        | 2.53        |
|                    | Campaign-date-mean                                 | -0.012       | 2.86        |
| Blocking unit      | Random $K$ -fold                                   | 0.596        | 1.81        |
|                    | By station                                         | 0.596        | 1.81        |
|                    | <b>By campaign date</b>                            | <b>0.574</b> | <b>1.86</b> |
|                    | By year                                            | 0.495        | 2.02        |
|                    | One record per event ( $n = 540$ )                 | 0.571        | 1.77        |
| Zone extrapolation | Leave out Lago Menor ( $n = 133$ )                 | 0.247        | 1.90        |
|                    | Leave out Bahía de Puno ( $n = 88$ )               | 0.036        | 2.11        |
|                    | Leave out Lago Mayor ( $n = 591$ )                 | -0.047       | 2.52        |
| Temporal           | Train $\leq 2019$ , test 2022 & 2024 ( $n = 226$ ) | 0.507        | 2.63        |

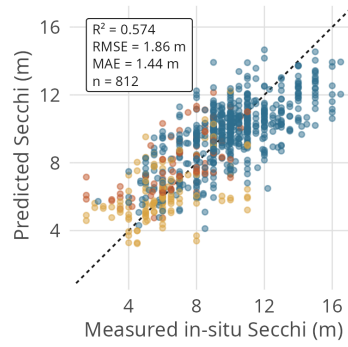
282 observed maximum of 16.5 m, which alone drives  $R^2$  to -2.458 and RMSE to  
283 5.29 m (Fig. 3d). Benchmarks of classical algorithms reported without phys-  
284 ical bounds should be treated with caution: a single unconstrained extrapo-  
285 lation manufactures an arbitrarily large advantage for the machine-learning  
286 model.

### 287 3.4. *Model family and tuning do not change the picture*

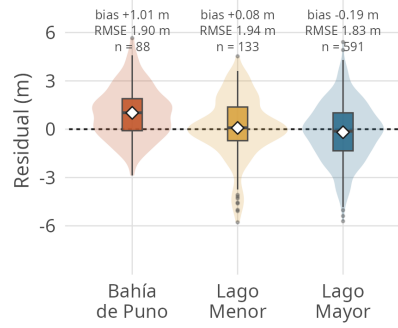
288 We compared four learner families, each tuned by Optuna inside nested  
289 cross-validation, against target transformations and reduced feature sets (Ta-  
290 ble A.5). The best configuration — an equal-weight ensemble of the four  
291 tuned models — reached  $R^2 = 0.583$ , an improvement of 0.009 over the  
292 untuned Random Forest. That gain does exceed the spread across random  
293 seeds (0.003 in  $R^2$ ), so it is not noise; but it corresponds to 0.02 m of RMSE  
294 on a lake whose reference measurement is recorded to the nearest half metre,  
295 and it changes no conclusion in this article. Neither log-transforming the  
296 target nor tuning individual learners helped more than that, and restricting  
297 the feature set cost accuracy.

298 We therefore retain the untuned Random Forest as the reported model:  
299 it is simpler, faster and easier to reproduce, and reporting a tuned ensemble  
300 for a 0.009 gain would overstate the precision of the product. The ceiling  
301 here is set by the data — a dry-season-only record, a reference measurement  
302 quantized at half a metre, and a target whose variance is dominated by a  
303 three-level trophic gradient — not by the choice of learner.

**(a) Predicted vs measured**  
Random Forest, campaign-blocked  
validation. Dashed line is 1:1

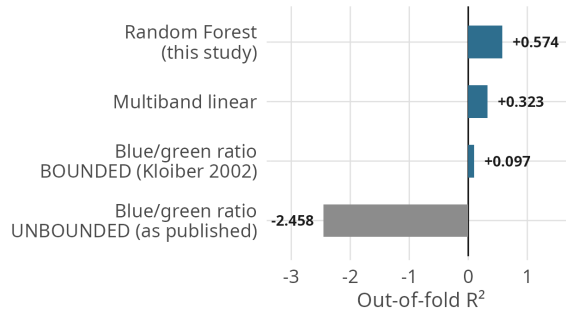


**(b) Bias concentrates in the bay**  
Residuals (predicted - measured) by zone.  
Diamond is the mean. Bahía de Puno, the  
most eutrophic zone, +1.0 m



• Bahía de Puno • Lago Menor • Lago Mayor

**(c) The baseline was misimplemented**  
Same validation design for all.  
The published baseline undid the log without  
bounds; a single 143 m prediction ruined its  $R^2$



**(d) One impossible value ruins  $R^2$**   
Classical ratio on log scale.  
The 143 m point is the one that produced  
the  $R^2 = -2.46$

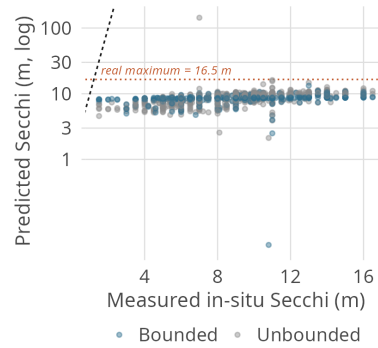


Figure 3: Retrieval under the primary design. (a) Out-of-fold predicted versus measured Secchi depth. (b) Residuals by trophic zone; the retrieval overestimates in eutrophic Bahía de Puno. (c) Benchmark against classical and linear baselines. (d) The unbounded classical implementation in log scale: one impossible prediction destroys the metric.

### 304 3.5. Where the retrieval stops working

305 Withholding an entire trophic zone collapses the retrieval (Table 1):  $R^2$   
306 falls to 0.247 for Lago Menor, 0.036 for Bahía de Puno and -0.047 for Lago  
307 Mayor. The model interpolates within optical regimes it has seen and does  
308 not extrapolate to a water type absent from training. On the temporal hold-  
309 out (train  $\leq 2019$ , test 2022 and 2024) it retains  $R^2 = 0.507$ , though 57 of the  
310 69 test stations also appear in training (82.6%), so this measures temporal  
311 rather than fully independent generalization.

312 Error is unevenly distributed (Table A.6). The retrieval carries a positive  
313 bias of 1.01 m in Bahía de Puno, the most eutrophic and most vulnerable  
314 zone, because the clear-water majority pulls predictions upward.

315 Two features of the harmonized archive should be read together with these  
316 numbers. Removing the two bands most corrupted by the land-derived in-  
317 tercepts lowers  $R^2$  only from 0.574 to 0.541; using scale-invariant ratios alone  
318 gives 0.468; and applying our water-specific recalibration gives 0.518. Train-  
319 ing on each sensor separately gives 0.563 (Sentinel-2,  $n = 274$ ) and 0.549  
320 (Landsat,  $n = 538$ ). The pooled model’s small advantage over these vari-  
321 ants is consistent with it exploiting sensor identity as an additional covariate  
322 rather than with genuine cross-sensor synergy.

### 323 3.6. Calibrated per-prediction uncertainty

324 Split-conformal intervals are well calibrated *marginally* at every nominal  
325 level tested: empirical coverage is 50.9% at 50% nominal, 79.1% at 80%,  
326 89.4% at 90% and 94.3% at 95%, a maximum calibration error of 0.94 per-  
327 centage points (Fig. A.7a).

328 They are not calibrated *conditionally*, and this matters more for opera-  
 329 tional use than the marginal result. Coverage falls to 71.3% in the clearest  
 330 water ( $Z_{SD} > 12.5$  m) and 79.5% in the most turbid ( $Z_{SD} < 5$  m), while  
 331 exceeding the nominal level in the middle of the range (Fig. A.7c). The  
 332 mechanism is structural rather than incidental: split conformal derives a sin-  
 333 gle global quantile of absolute residuals, so the interval has constant width  
 334 (6.12 m at 90%) everywhere and cannot adapt to the local difficulty of the re-  
 335 trieval. It therefore buys its marginal guarantee by over-covering in the well-  
 336 sampled middle of the range and under-covering at both extremes — which  
 337 are precisely where management decisions are made. A locally adaptive vari-  
 338 ant such as conformalized quantile regression (Romano et al., 2019) would  
 339 be required for reliable conditional coverage, and we recommend reporting  
 340 coverage conditional on the predicted value as standard practice whenever  
 341 conformal intervals accompany an operational product.

342 For a lake whose transparency spans 1.5–16.5 m, a  $\pm 3$  m interval is in-  
 343 formative for zone-scale contrasts and multi-metre change, and is not infor-  
 344 mative for sub-metre differences.

### 345 3.7. Interpretation and retrievability

346 Out-of-fold SHAP attributes 72.6% of the retrieval to green and blue  
 347 bands and their ratios, 14.4% to NIR and SWIR, and the remainder to  
 348 other indices (Fig. 4a). Green/blue alone accounts for 37%. The dependence  
 349 is monotone and of the sign expected from water-clarity optics (Fig. 4b):  
 350 a higher green-to-blue ratio pushes the prediction toward less transparent  
 351 water. The retrieval is therefore grounded in optics rather than location  
 352 memorization — while resting on the band pair with the worst cross-sensor

353 agreement.

354 With real labels and the same campaign-blocked design, no other vari-  
355 able is retrievable (Fig. 4c): chlorophyll-*a* reaches  $R^2 = 0.012$ , suspended  
356 solids 0.056, in-situ water temperature 0.113, and Landsat thermal surface  
357 temperature 0.020 ( $r = 0.14$ ,  $n = 662$ ). Their in-situ ranges — chlorophyll-*a*  
358 0–4.6 mg m<sup>-3</sup>, suspended solids mostly below 10 mg L<sup>-1</sup> — lie below the op-  
359 tical and thermal detection limits of these sensors. We subject the negative  
360 claim to the same design as the positive one, and report it to caution against  
361 applying chlorophyll algorithms (Gitelson et al., 2009; Mishra and Mishra,  
362 2012) or thermal retrievals (Ruiz-Verdú et al., 2016) in clear high-altitude  
363 water without local validation.

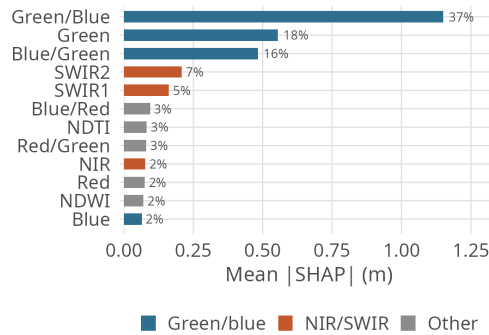
### 364 3.8. Temporal patterns and the fragility of trend claims

365 Mann–Kendall tests on annual median in-situ transparency over the full  
366 2011–2024 record show no statistically significant trend in any zone (Bahía  
367 de Puno: Sen slope 0.16 m yr<sup>-1</sup>,  $p = 0.241$ ; Lago Mayor: 0.25 m yr<sup>-1</sup>,  
368  $p = 0.138$ ; Fig. A.8a).

369 That conclusion is fragile in a way worth reporting explicitly. Restricting  
370 the series to 2013 onwards — the natural choice if trends were computed  
371 on the satellite match-up subset, whose record begins with Landsat 8 —  
372 turns both Bahía de Puno ( $p = 0.017$ ) and Lago Mayor (Sen +0.42 m yr<sup>-1</sup>,  
373  $p = 0.012$ ) significantly positive. Across start years from 2011 to 2014, 5 of  
374 12 zone–start-year combinations change verdict (Fig. A.8b). The year 2013  
375 has the lowest annual medians on record, so beginning there manufactures  
376 an upward trend. With 9–11 annual values and three gap years the test has  
377 low power, and the correct reading is absence of evidence, not evidence of

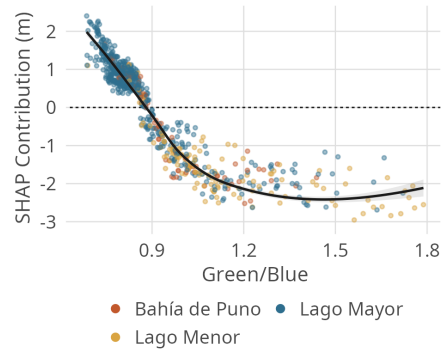
**(a) Green and blue dominate attribution**

Mean |SHAP| computed OUT-OF-FOLD: each observation is explained by the model that did not see it. Labels are the share of the total



**(b) How Green/Blue acts**

SHAP value vs. feature value. Greener water relative to blue pushes the prediction toward less transparent water. Trimmed to the 1-99th percentile (18 extreme points out)



**(c) Only transparency is retrievable in this water**

Random Forest with the same honest design (campaign blocking) for each variable. The negative claim is subject to the same rigor as the positive one

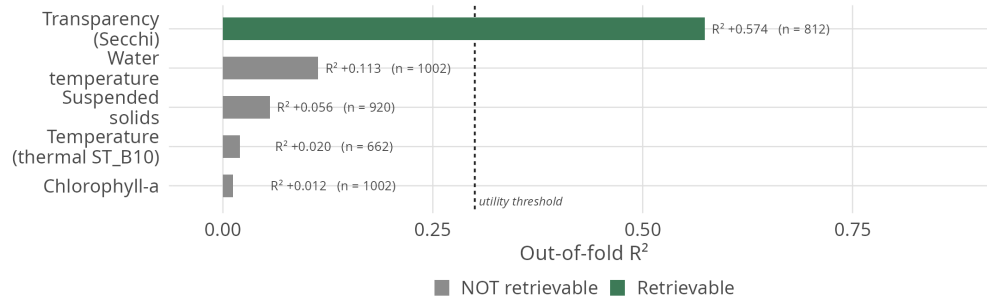


Figure 4: Interpretation and operational envelope. (a) Out-of-fold SHAP importance; green and blue dominate. (b) SHAP dependence on the leading feature, by trophic zone. (c) Retrievability of each water-quality variable under the same campaign-blocked design; only transparency clears a useful threshold.

378 stability. We recommend that trend analyses in campaign-based lake records  
379 report this sensitivity as a matter of course.

## 380 4. Discussion

### 381 4.1. Implications for harmonized multi-mission archives

382 The virtual-constellation approach (Claverie et al., 2018; Pahlevan et al.,  
383 2019) has become the default for multi-decadal lake monitoring, and the Roy  
384 et al. (2016) coefficients are its most reused transform. Our results place  
385 a boundary on where that default is safe. Those coefficients were fitted on  
386 terrestrial targets and carry additive intercepts appropriate to surfaces one  
387 to two orders of magnitude brighter than clear water. Over Lake Titicaca the  
388 intercept exceeds the native over-water signal by a factor of 69.8 in the near-  
389 infrared and by 2.5 even in blue, and the two sensors remain 99.8% separable  
390 after the transform. That water reflectance from these products is unreliable  
391 is already documented (Pahlevan et al., 2019; Masek et al., 2021); what our  
392 paired observations add is the magnitude — the problem is not confined  
393 to the near-infrared, and it is large enough to make the harmonized bands  
394 function as sensor labels for a machine-learning model — and the finding that  
395 local recalibration repairs the aggregate spectral signature without repairing  
396 per-observation agreement in blue.

397 The practical consequence is direct. A machine-learning model trained  
398 on a pooled, land-harmonized archive can achieve apparently good accuracy  
399 while having learned two sensor-specific mappings joined by a readable sensor  
400 label. That is not a harmonized retrieval, and it will not behave as one when  
401 applied to a scene from either sensor alone. The diagnostic is cheap and we

402 recommend it as routine: train a classifier to predict the source sensor from  
403 the harmonized features and report its accuracy against the majority-class  
404 baseline. If the sensors remain separable, the harmonization has not done  
405 its job. For clear oligotrophic lakes we further recommend deriving water-  
406 specific coefficients from same-day paired observations, reporting per-band  
407 agreement, and — where blue-band agreement is as poor as we find here —  
408 reporting single-sensor retrievals alongside any pooled product.

#### 409 *4.2. Implications for validation design*

410 Blocking by station is the standard remedy for spatial autocorrelation  
411 in this literature, and in our data it changes nothing: random and station-  
412 blocked folds give identical accuracy. Read conventionally, that would be  
413 reassuring. The null models show it is not. Under campaign-blocked folds  
414 the station-mean null still explains 21% of variance and the zone-mean null  
415 29%, while the campaign-date-mean null explains none. In a campaign-based  
416 monitoring programme the whole lake is sampled within a few days under  
417 shared atmospheric and limnological conditions, so observations cluster by  
418 campaign far more strongly than by location; station-blocked folds leave that  
419 structure entirely intact.

420 The general point is that the blocking unit should be diagnosed, not  
421 assumed. Reporting null models under each candidate design costs little  
422 and identifies the unit that carries the dependence. This applies well beyond  
423 Lake Titicaca: most large-lake monitoring programmes sample in campaigns,  
424 so the same mismatch between the conventional blocking unit and the real  
425 dependence structure is likely to be widespread.

426 For our retrieval, the honest figure is  $R^2 = 0.574$  (RMSE 1.86 m) under

427 campaign-blocked validation, against 0.287 for a zone-mean null: the model  
428 roughly doubles the skill available from knowing only which basin a sample  
429 came from. That is a real but modest contribution, and stating it that way  
430 is more useful to a manager than an unqualified  $R^2$ .

#### 431 4.3. Operational envelope for Lake Titicaca

432 Our RMSE of 1.86 m spans roughly 12% of the observed  $Z_{SD}$  range  
433 and sits below the published range for  $Z_{SD}$  retrieval ( $R^2 \approx 0.6$ – $0.9$ ; Zhang  
434 et al., 2022; Lee et al., 2015), obtained here under a predominantly clear  
435 oligotrophic regime and a stricter validation design. It is also approaching  
436 the granularity of the reference measurement itself: disk readings are recorded  
437 to the nearest half metre, giving only 73 distinct values across 812 match-ups.

438 Four constraints bound operational use. The retrieval carries a 1.01 m  
439 positive bias in Bahía de Puno — the most eutrophic and most vulnerable  
440 zone — so mapped values there must be read with their intervals and con-  
441 firmed by field sampling. Leave-one-zone-out results show the model cannot  
442 extrapolate to an unsampled optical regime; integration must be incremen-  
443 tal, each new campaign both validating the current map and extending the  
444 training set. The retrieval is calibrated and validated for the austral dry sea-  
445 son only, and the reason is worth stating precisely because it determines the  
446 remedy. It is not that wet-season imagery is unusable at 3810 m, nor that our  
447 match-up criteria excluded it: it is that the field programme has produced  
448 a single wet-season campaign in fourteen years, and that campaign predates  
449 both satellites. Wet-season blooms in Bahía de Puno — the phenomenon a  
450 transparency product would most usefully warn about — are therefore en-  
451 tirely unrepresented, and the model should not be applied between December

452 and March.

453 This converts the limitation into a specific, actionable recommendation  
454 for the monitoring authorities. Extending the protocol to the wet season does  
455 not require a better algorithm; it requires field data that do not yet exist. A  
456 single additional wet-season campaign per year, timed to any cloud-free win-  
457 dow and covering the three trophic zones, would within a few years supply  
458 enough independent campaign-blocks to calibrate and validate a wet-season  
459 retrieval under the same design used here. Because the binding constraint is  
460 sampling rather than sensing, this is among the cheapest available improve-  
461 ments to Lake Titicaca’s water-quality monitoring capability. And because  
462 conformal coverage degrades at both extremes of the transparency range, the  
463 nominal 90% interval should be treated as optimistic in the clearest and most  
464 turbid water.

465 Within those bounds the product is useful to the regional monitoring  
466 authorities (IMARPE, ALT, OEFA, and the binational management frame-  
467 work). A calibrated  $\pm 3$  m interval resolves zone-scale contrasts and multi-  
468 metre change, which is the scale at which clarity loss in Bahía de Puno  
469 would be actionable, and a single transboundary product under one cali-  
470 bration supports comparable assessment across the Peru–Bolivia boundary  
471 where coordinated field monitoring is logistically difficult. It does not resolve  
472 sub-metre change and should not be presented as doing so.

#### 473 4.4. *Limitations*

474 Beyond the constraints already stated, three limitations bound this work.  
475 The in-situ record is episodic, with no campaigns in 2020, 2021, 2023 and only  
476 one wet-season campaign in fourteen years, so the temporal hold-out rests

477 on two test years, the trend analysis on 9–11 annual values, and the seasonal  
478 envelope on none. Transparency above approximately 15 m saturates the  
479 optical signal. And this is a single lake: the two diagnostics we propose are  
480 general, but the water-specific coefficients in Table A.3 are local to Titicaca  
481 and must be refitted elsewhere.

## 482 5. Conclusions

483 Using 1002 satellite–field match-ups over twelve years, we show that two  
484 practices imported from the turbid-water remote-sensing literature fail in a  
485 clear high-altitude large lake. Cross-sensor harmonization coefficients derived  
486 over land leave Sentinel-2 and Landsat 99.8% separable over oligotrophic  
487 water, because their additive intercepts exceed the native over-water signal  
488 by up to a factor of 69.8; and validation blocked by monitoring station leaves  
489 the dominant dependence — between campaigns — entirely intact.

490 Correcting both, a Random Forest retrieves Secchi depth with  $R^2 = 0.574$   
491 and RMSE 1.86 m, roughly double the skill of a zone-mean null, with confor-  
492 mal prediction intervals that are calibrated marginally but under-cover at the  
493 extremes of the transparency range. Skill collapses under extrapolation to an  
494 unsampled trophic zone; chlorophyll-*a*, suspended solids and thermal tem-  
495 perature are not retrievable at all; and no transparency trend is significant  
496 once the full in-situ record is used rather than the satellite-era subset.

497 We therefore recommend two routine checks for multi-mission lake water-  
498 quality products, neither of which requires new methodology: report the  
499 accuracy of a classifier two-sample test on the harmonized features, and re-  
500 port null models under each candidate validation blocking unit. Both cost

one model fit, and both distinguish genuine optical retrieval from sensor labelling and sampling design. For Lake Titicaca the resulting product is a dry-season, zone-scale transparency indicator with honest  $\pm 3$  m intervals — less than has sometimes been claimed for such lakes, and more than the region has had.

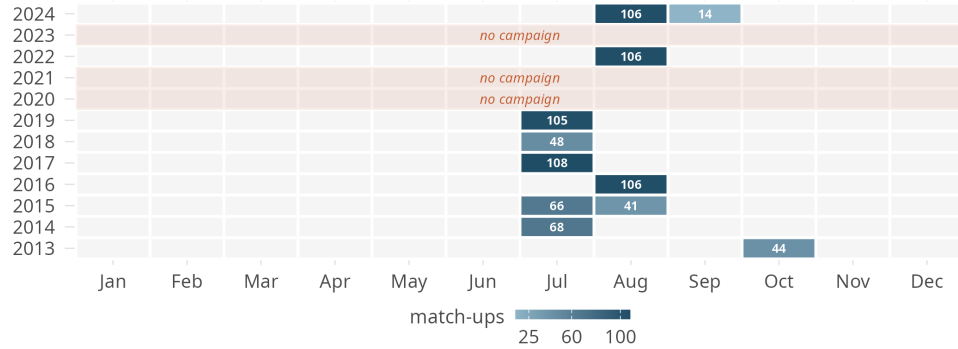
## Appendix A. Supplementary Figures and Tables

Table A.2: Effect of the Roy et al. (2016) OLI→MSI transform over Lake Titicaca water. Native OLI is the Collection-2 surface reflectance recovered by inverting the transform. Reflectance is dimensionless.

| Band  | Roy intercept $a$ | Native OLI<br>median | Intercept /<br>signal | Harmonized<br>LS median | Harmonized /<br>S2 median |
|-------|-------------------|----------------------|-----------------------|-------------------------|---------------------------|
| Blue  | 0.0183            | +0.0072              | 2.5                   | 0.0247                  | 1.74                      |
| Green | 0.0123            | +0.0089              | 1.4                   | 0.0206                  | 1.61                      |
| Red   | 0.0123            | +0.0013              | 10.6                  | 0.0134                  | 2.66                      |
| NIR   | 0.0448            | −0.0006              | 69.8                  | 0.0443                  | 15.42                     |
| SWIR1 | 0.0306            | +0.0013              | 23.2                  | 0.0317                  | 13.34                     |
| SWIR2 | 0.0116            | +0.0013              | 8.2                   | 0.0129                  | 5.70                      |

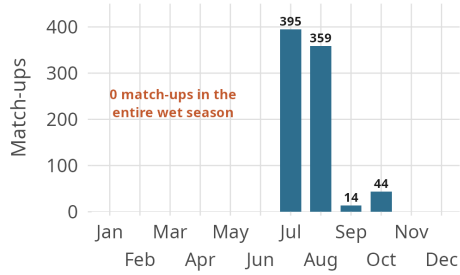
**(a) Three years with no campaigns and no data outside the dry season**

Number of satellite-field match-ups by year and month. There is no data for 2020, 2021 or 2023, so the 2022-2024 hold-out only contains 2022 and 2024



**(b) Only dry season data exists**

Bahía de Puno blooms occur during the wet season, which the dataset does not cover at all



■ Wet season (Dec-Mar) ■ Transition ■ Dry season (Jul-Oct)

**(c) The trophic gradient dominates Secchi**

Only 73 distinct values in 812 match-ups: readings are rounded to half a metre, so the RMSE of 1.86 m borders the granularity of the reference measurement itself

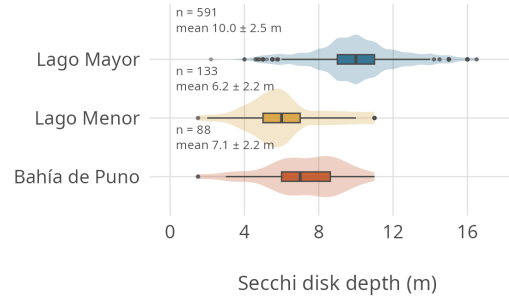


Figure A.5: What the dataset actually contains. (a) Match-ups by year and month; three years have no campaign at all, so a hold-out described as “2022–2024” contains only 2022 and 2024. (b) Seasonal coverage: zero match-ups in the wet season, when blooms occur in Bahía de Puno. (c) Secchi distribution by trophic zone; the reference measurement is quantized at roughly half a metre.

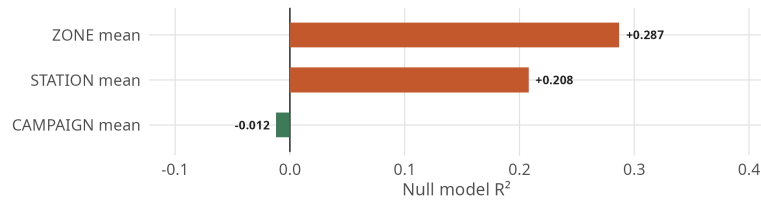
Table A.3: Water-specific LS→S2 recalibration fitted on 270 paired station–date observations, of the form  $\rho_{S2} = a + b \rho_{LS,Roy}$ .

| Band  | $a$     | $b$   | Pearson $r$ | Agreement  |
|-------|---------|-------|-------------|------------|
| Blue  | +0.0084 | 0.258 | 0.26        | unusable   |
| Green | −0.0009 | 0.665 | 0.71        | good       |
| Red   | −0.0005 | 0.427 | 0.52        | weak       |
| NIR   | −0.0250 | 0.631 | 0.62        | acceptable |
| SWIR1 | −0.0223 | 0.770 | 0.64        | acceptable |
| SWIR2 | −0.0057 | 0.615 | 0.60        | acceptable |

**(a) Only campaign blocking neutralizes its own null model**

$R^2$  of models predicting only the group mean, with folds blocked by campaign.

A blocking unit is effective if and only if its own null model collapses to zero under it



**(b) Station blocking changes nothing; extrapolation breaks everything**

Out-of-fold  $R^2$  of the Random Forest; the large point is the primary design.

The fact that station blocking matches with random does not prove absence of leakage: it proves that it retains nothing

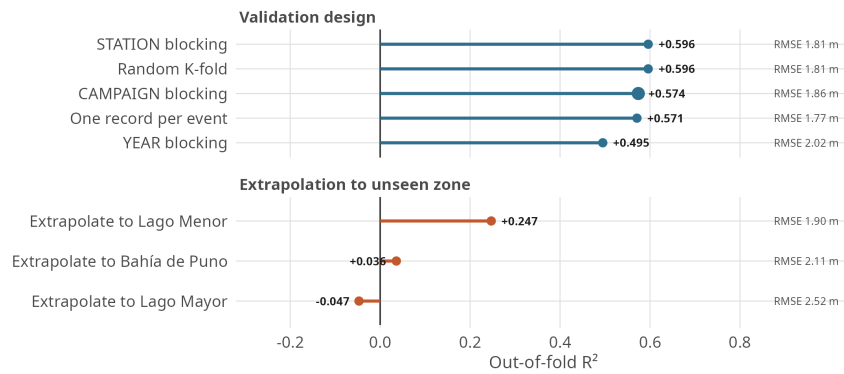
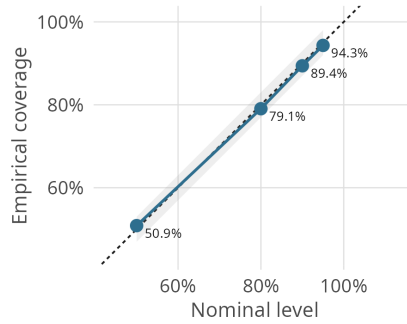
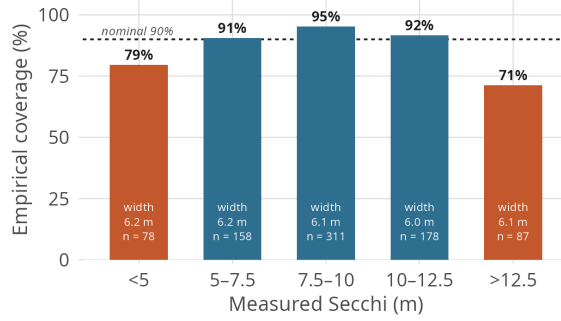


Figure A.6: Diagnosing the blocking unit. (a) Null models evaluated under campaign-blocked folds: only the campaign-mean null collapses, so only campaign blocking is doing work. (b) Out-of-fold  $R^2$  under each validation design and under extrapolation to an unseen trophic zone. That station blocking matches the random split is not evidence of no leakage; it is evidence that station blocking retains nothing.

**(a) Calibrated, on average**  
Out-of-fold empirical coverage vs nominal. Gray band is  $\pm 3$  points



**(c) But not at the ends of the range**  
Split-conformal uses a single global quantile, so the width is constant ( $\sim 6.1$  m) and the interval cannot adapt: it undercovers in the clearest water



**(b) The 90% interval vs the measurement**

Match-ups ordered by measured transparency. The line is the prediction, the band is the 90% conformal interval. 86 out of 812 points fall outside

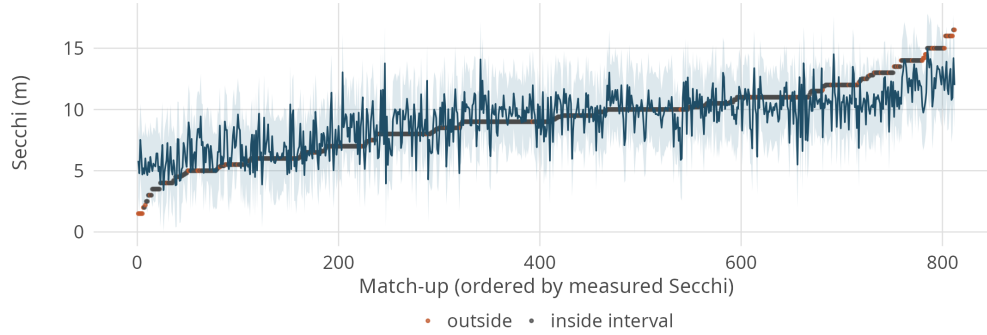
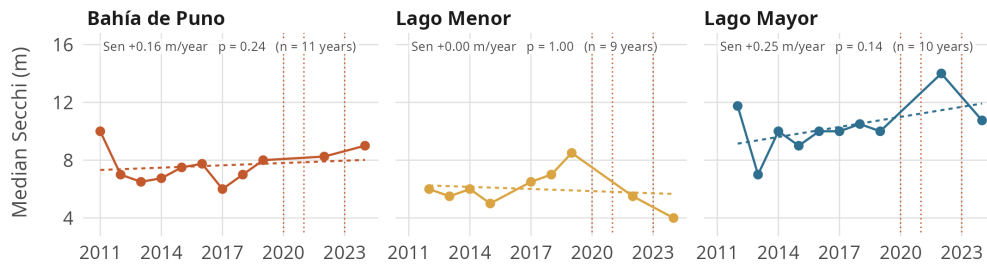


Figure A.7: Conformal prediction intervals under campaign-blocked validation. (a) Empirical versus nominal coverage: marginally calibrated across the 50–95% range. (b) The 90% interval against the measured value. (c) Coverage conditional on measured transparency. Because split conformal uses a single global residual quantile, the interval width is constant ( $\approx 6.12$  m) and cannot adapt, so the marginal guarantee is bought by over-covering in the middle of the range and under-covering at both extremes.

**(a) No zone shows significant trend in the full record**

Annual median transparency in-situ (2011–2024, 734 readings). The vertical dotted lines mark the years with no campaign: 2020, 2021 and 2023



**(b) The verdict is set by the start year, not the lake**

Same Mann-Kendall test, same data, different start year. Starting in 2013 — the year with the lowest medians in the record — turns two zones into "significantly more transparent"

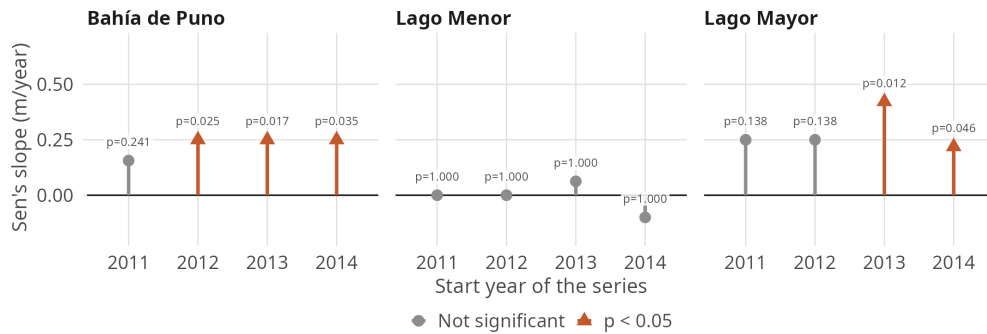


Figure A.8: Transparency trends and their fragility. (a) Annual median in-situ transparency by zone over the full 2011–2024 record, with Mann–Kendall/Sen statistics; no trend is significant. (b) The same test with different start years: beginning the series in 2013, the lowest-median year on record, turns two zones significantly positive.

Table A.4: Head-to-head benchmark, campaign-blocked GroupKFold,  $n = 812$ .

| Algorithm                                    | $R^2$        | RMSE        | MAE         | Bias  |
|----------------------------------------------|--------------|-------------|-------------|-------|
| Classical blue/green, unbounded <sup>a</sup> | −2.458       | 5.29        | 1.91        | −0.19 |
| Classical blue/green, bounded <sup>b</sup>   | 0.097        | 2.71        | 2.13        | −0.46 |
| Multiband linear                             | 0.323        | 2.34        | 1.78        | −0.01 |
| Random Forest (this study)                   | <b>0.574</b> | <b>1.86</b> | <b>1.44</b> | −0.02 |

RMSE, MAE and bias in metres.

<sup>a</sup> back-transformed without bounding predictions; see text.

<sup>b</sup> standard form of Kloiber et al. (2002).

Table A.5: Model selection under nested cross-validation (Optuna inside an inner campaign-blocked split of each outer training partition). Selected configurations; the full grid is in the repository.

| Model                                                                                | Features     | $R^2$        | RMSE (m)    |
|--------------------------------------------------------------------------------------|--------------|--------------|-------------|
| Ensemble of the four tuned models                                                    | all 12       | 0.583        | 1.84        |
| LightGBM, tuned                                                                      | all 12       | 0.581        | 1.84        |
| XGBoost, tuned                                                                       | all 12       | 0.577        | 1.85        |
| Random Forest, tuned                                                                 | all 12       | 0.576        | 1.85        |
| <b>Random Forest, untuned (reported)</b>                                             | all 12       | <b>0.574</b> | <b>1.86</b> |
| Extremely Randomized Trees, untuned                                                  | all 12       | 0.573        | 1.86        |
| XGBoost, tuned, $\log Z_{SD}$                                                        | all 12       | 0.561        | 1.89        |
| LightGBM, tuned                                                                      | visible only | 0.544        | 1.92        |
| LightGBM, tuned                                                                      | ratios only  | 0.488        | 2.04        |
| Seed-to-seed spread of the untuned Random Forest: $R^2$ range 0.003 over five seeds. |              |              |             |

Table A.6: Out-of-fold error by trophic zone, campaign-blocked validation.

| Zone          | $R^2$ | RMSE (m) | Bias (m) | $n$ |
|---------------|-------|----------|----------|-----|
| Bahía de Puno | 0.221 | 1.90     | +1.01    | 88  |
| Lago Menor    | 0.211 | 1.94     | +0.08    | 133 |
| Lago Mayor    | 0.448 | 1.83     | −0.19    | 591 |

507 **CRedit authorship contribution statement**

508 **Dina Maribel Yana-Yucra:** Conceptualization, Data curation, For-  
509 mal analysis, Investigation, Methodology, Project administration, Resources,  
510 Software, Supervision, Validation, Visualization, Writing – original draft,  
511 Writing – review & editing. **Richar Andre Vilca-Solorzano:** Conceptual-  
512 ization, Data curation, Formal analysis, Investigation, Methodology, Project  
513 administration, Resources, Software, Validation, Visualization, Writing –  
514 original draft, Writing – review & editing. **Fred Torres-Cruz:** Conceptual-  
515 ization, Data curation, Formal analysis, Investigation, Methodology, Project  
516 administration, Resources, Supervision, Validation, Visualization, Writing –  
517 original draft, Writing – review & editing. **Vladimiro Ibañez:** Conceptual-  
518 ization, Data curation, Formal analysis, Investigation, Methodology, Project  
519 administration, Resources, Supervision, Validation, Visualization, Writing –  
520 original draft, Writing – review & editing.

521 All four authors contributed substantively across every stage of the study,  
522 and all read and approved the final manuscript.

523 **Declaration of competing interest**

524 The authors declare that they have no known competing financial inter-  
525 ests or personal relationships that could have appeared to influence the work  
526 reported in this paper.

## 527 **Data availability**

The satellite–field match-up datasets and the complete analysis pipeline, including the scripts that generate every number, table and figure in this article, are openly available at:

528 <https://github.com/Andre031222/harmonized-titicaca-transparency>



Repository

Raw imagery is freely available via Google Earth Engine; the in-situ measurements derive from published IMARPE/ALT and OEFA monitoring records.

## 529 **Acknowledgements**

530 We thank IMARPE, ALT and OEFA for the publicly available monitoring  
531 records that made this study possible. This research received no external  
532 funding.

## 533 **References**

- 534 Akiba, T., Sano, S., Yanase, T., Ohta, T., Koyama, M., 2019. Optuna: A  
535 next-generation hyperparameter optimization framework, in: Proceedings  
536 of the 25th ACM SIGKDD International Conference on Knowledge Dis-  
537 covery and Data Mining, pp. 2623–2631. doi:10.1145/3292500.3330701.
- 538 Angelopoulos, A.N., Bates, S., 2023. Conformal prediction: A gentle in-  
539 troduction. Foundations and Trends in Machine Learning 16, 494–591.  
540 doi:10.1561/22000000101.

541 Ayala Izurieta, J.E., Beltrán Dávalos, A.A., Jara Santillán, C.A.,  
542 Godoy Ponce, S.C., Van Wittenberghe, S., Verrelst, J., Delegido, J.,  
543 2023. Spatial and temporal analysis of water quality in high Andean  
544 lakes with Sentinel-2 satellite automatic water products. *Sensors* 23, 8774.  
545 doi:10.3390/s23218774.

546 Breiman, L., 2001. Random forests. *Machine Learning* 45, 5–32. doi:10.102  
547 3/A:1010933404324.

548 Cao, Z., Ma, R., Liu, M., Duan, H., Xiao, Q., Xue, K., Shen, M., 2022.  
549 Harmonized chlorophyll-a retrievals in inland lakes from Landsat-8/9 and  
550 Sentinel 2A/B virtual constellation through machine learning. *IEEE Trans-*  
551 *actions on Geoscience and Remote Sensing* 60, 1–16. doi:10.1109/TGRS.2  
552 022.3207345.

553 Chen, T., Guestrin, C., 2016. XGBoost: A scalable tree boosting system,  
554 in: *Proceedings of the 22nd ACM SIGKDD International Conference on*  
555 *Knowledge Discovery and Data Mining*, pp. 785–794. doi:10.1145/2939  
556 672.2939785.

557 Claverie, M., Ju, J., Masek, J.G., Dungan, J.L., Vermote, E.F., Roger, J.C.,  
558 Skakun, S.V., Justice, C., 2018. The harmonized landsat and Sentinel-2  
559 surface reflectance data set. *Remote Sensing of Environment* 219, 145–161.  
560 doi:10.1016/j.rse.2018.09.002.

561 Friedman, J.H., 2003. On multivariate goodness-of-fit and two-sample test-  
562 ing. *Statistical Problems in Particle Physics, Astrophysics and Cosmology*  
563 (PHYSTAT) , 311–313.

- 564 Gholizadeh, M.H., Melesse, A.M., Reddi, L., 2016. A comprehensive review  
565 on water quality parameters estimation using remote sensing techniques.  
566 Sensors 16, 1298. doi:10.3390/s16081298.
- 567 Gitelson, A.A., Gurlin, D., Moses, W.J., Barrow, T., 2009. A bio-optical  
568 algorithm for the remote estimation of the chlorophyll-a concentration in  
569 case 2 waters. Environmental Research Letters 4, 045003. doi:10.1088/  
570 1748-9326/4/4/045003.
- 571 Gorelick, N., Hancher, M., Dixon, M., Ilyushchenko, S., Thau, D., Moore,  
572 R., 2017. Google Earth Engine: Planetary-scale geospatial analysis for  
573 everyone. Remote Sensing of Environment 202, 18–27. doi:10.1016/j.rs  
574 e.2017.06.031.
- 575 Heredia, C., Guédron, S., Point, D., Perrot, V., Campillo, S., Verin, C., Es-  
576 pinoza, M.E., Fernandez, P., Duwig, C., Achá, D., 2022. Anthropogenic  
577 eutrophication of Lake Titicaca (Bolivia) revealed by carbon and nitro-  
578 gen stable isotopes fingerprinting. Science of the Total Environment 845,  
579 157286. doi:10.1016/j.scitotenv.2022.157286.
- 580 Kloiber, S.M., Brezonik, P.L., Olmanson, L.G., Bauer, M.E., 2002. A proce-  
581 dure for regional lake water clarity assessment using Landsat multispectral  
582 data. Remote Sensing of Environment 82, 38–47. doi:10.1016/S0034-4  
583 257(02)00022-6.
- 584 Lanza, W.G., Hernández, V.C., Achá, D., Lazzaro, X., 2024. Responses  
585 of phytoplankton and periphyton community structure to an anthropic

586 eutrophication gradient in tropical high-altitude Lake Titicaca. *Journal of*  
587 *Great Lakes Research* 50, 102294. doi:10.1016/j.jglr.2024.102294.

588 Lee, Z., Shang, S., Hu, C., Du, K., Weidemann, A., Hou, W., Lin, J., Lin,  
589 G., 2015. Secchi disk depth: A new theory and mechanistic model for  
590 underwater visibility. *Remote Sensing of Environment* 169, 139–149. doi:10  
591 .1016/j.rse.2015.08.002.

592 Lopez-Paz, D., Oquab, M., 2017. Revisiting classifier two-sample tests, in:  
593 *International Conference on Learning Representations (ICLR)*. URL: <https://openreview.net/forum?id=SJkXfE5xx>.  
594 <https://openreview.net/forum?id=SJkXfE5xx>.

595 Lundberg, S.M., Erion, G., Chen, H., DeGrave, A., Prutkin, J.M., Nair,  
596 B., Katz, R., Himmelfarb, J., Bansal, N., Lee, S.I., 2020. From local  
597 explanations to global understanding with explainable AI for trees. *Nature*  
598 *Machine Intelligence* 2, 56–67. doi:10.1038/s42256-019-0138-9.

599 Lundberg, S.M., Lee, S.I., 2017. A unified approach to interpreting model  
600 predictions, in: *Advances in Neural Information Processing Systems* 30  
601 (NeurIPS 2017), pp. 4765–4774. URL: [https://proceedings.neurips.](https://proceedings.neurips.cc/paper/2017/hash/8a20a8621978632d76c43dfd28b67767-Abstract.html)  
602 [cc/paper/2017/hash/8a20a8621978632d76c43dfd28b67767-Abstract.](https://proceedings.neurips.cc/paper/2017/hash/8a20a8621978632d76c43dfd28b67767-Abstract.html)  
603 [html](https://proceedings.neurips.cc/paper/2017/hash/8a20a8621978632d76c43dfd28b67767-Abstract.html).

604 Maligaya, V.H., Baltodano, A., Agramont, A., van Griensven, A., 2024. Ex-  
605 ploring trends and variability of water quality over Lake Titicaca using  
606 global remote sensing products. *Remote Sensing* 16, 4785. doi:10.3390/  
607 rs16244785.

608 Mann, H.B., 1945. Nonparametric tests against trend. *Econometrica* 13,  
609 245–259. doi:10.2307/1907187.

610 Masek, J.G., Ju, J., Roger, J.C., Skakun, S., Vermote, E., Claverie, M.,  
611 Dungan, J., Yin, Z., Freitag, B., Justice, C., 2021. HLS Harmonized  
612 Landsat Sentinel-2 Product User Guide, Version 2.0. Technical Report.  
613 NASA Goddard Space Flight Center / Land Processes DAAC.

614 Mishra, S., Mishra, D.R., 2012. Normalized difference chlorophyll index: A  
615 novel model for remote estimation of chlorophyll-a concentration in turbid  
616 productive waters. *Remote Sensing of Environment* 117, 394–406. doi:10  
617 .1016/j.rse.2011.10.016.

618 OEFA, 2016. Informe 076-2016-OEFA/DE-SDCA: Evaluación de la calidad  
619 ambiental de la Bahía Interior de Puno, lago Titicaca. Technical Report.  
620 Organismo de Evaluación y Fiscalización Ambiental, Perú.

621 Pahlevan, N., Chittimalli, S.K., Balasubramanian, S.V., Vellucci, V., 2019.  
622 Sentinel-2/Landsat-8 product consistency and implications for monitoring  
623 aquatic systems. *Remote Sensing of Environment* 220, 19–29. doi:10.101  
624 6/j.rse.2018.10.027.

625 Pahlevan, N., Smith, B., Schalles, J., Binding, C., Cao, Z., Ma, R., Alikas,  
626 K., Kangro, K., Gurlin, D., Hà, N., Matsushita, B., Moses, W., Greb, S.,  
627 Lehmann, M.K., Ondrusek, M., Oppelt, N., Stumpf, R., 2020. Seamless  
628 retrievals of chlorophyll-a from Sentinel-2 (MSI) and Sentinel-3 (OLCI) in  
629 inland and coastal waters: A machine-learning approach. *Remote Sensing*  
630 of *Environment* 240, 111604. doi:10.1016/j.rse.2019.111604.

- 631 Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel,  
632 O., Blondel, M., Prettenhofer, P., Weiss, R., Dubourg, V., Vanderplas, J.,  
633 Passos, A., Cournapeau, D., Brucher, M., Perrot, M., Duchesnay, É., 2011.  
634 Scikit-learn: Machine learning in Python. *Journal of Machine Learning*  
635 *Research* 12, 2825–2830. URL: [https://jmlr.org/papers/v12/pedreg](https://jmlr.org/papers/v12/pedregosa11a.html)  
636 [osa11a.html](https://jmlr.org/papers/v12/pedregosa11a.html).
- 637 Roberts, D.R., Bahn, V., Ciuti, S., Boyce, M.S., Elith, J., Guillerá-Arroita,  
638 G., Hauenstein, S., Lahoz-Monfort, J.J., Schröder, B., Thuiller, W.,  
639 Warton, D.I., Wintle, B.A., Hartig, F., Dormann, C.F., 2017. Cross-  
640 validation strategies for data with temporal, spatial, hierarchical, or phy-  
641 logenetic structure. *Ecography* 40, 913–929. doi:10.1111/ecog.02881.
- 642 Romano, Y., Patterson, E., Candès, E.J., 2019. Conformalized quantile  
643 regression, in: *Advances in Neural Information Processing Systems* 32  
644 (NeurIPS 2019), pp. 3543–3553. URL: [https://proceedings.neurip](https://proceedings.neurips.cc/paper/2019/hash/5103c3584b063c431bd1268e9b5e76fb-Abstract.html)  
645 [s.cc/paper/2019/hash/5103c3584b063c431bd1268e9b5e76fb-Abstrac](https://proceedings.neurips.cc/paper/2019/hash/5103c3584b063c431bd1268e9b5e76fb-Abstract.html)  
646 [t.html](https://proceedings.neurips.cc/paper/2019/hash/5103c3584b063c431bd1268e9b5e76fb-Abstract.html).
- 647 Roy, D.P., Kovalsky, V., Zhang, H.K., Vermote, E.F., Yan, L., Kumar, S.S.,  
648 Egorov, A., 2016. Characterization of Landsat-7 to Landsat-8 reflective  
649 wavelength and normalized difference vegetation index continuity. *Remote*  
650 *Sensing of Environment* 185, 57–70. doi:10.1016/j.rse.2015.12.024.
- 651 Ruiz-Verdú, A., Jiménez, J.C., Lazzaro, X., Tenjo, C., Delegido, J., Pereira,  
652 M., Sobrino, J.A., Moreno, J.F., 2016. Comparison of MODIS and  
653 Landsat-8 retrievals of chlorophyll-a and water temperature over Lake Tit-

654     icaca, in: 2016 IEEE International Geoscience and Remote Sensing Sym-  
655     posium (IGARSS), pp. 7643–7646. doi:10.1109/IGARSS.2016.7730993.

656     Sen, P.K., 1968. Estimates of the regression coefficient based on Kendall’s  
657     tau. *Journal of the American Statistical Association* 63, 1379–1389. doi:10  
658     .1080/01621459.1968.10480934.

659     Siguayro Mamani, H., Franco, Y.B., 2022. Aspectos fisicoquímicos del lago  
660     Titicaca, julio 2019. *Informe Instituto del Mar del Perú* 49, 538–551.

661     Siguayro Mamani, H., Pasapera, J., Villanueva, C., Coila, Y., Gamarra, C.,  
662     2022. Evaluación de fuentes contaminantes en el anillo circunlacustre del  
663     lago Titicaca (sector peruano), 2017. *Boletín Instituto del Mar del Perú*  
664     37, 361–386. doi:10.53554/boletin.v37i2.378.

665     Topp, S.N., Pavelsky, T.M., Jensen, D., Simard, M., Ross, M.R.V., 2020.  
666     Research trends in the use of remote sensing for inland water quality  
667     science: Moving towards multidisciplinary applications. *Water* 12, 169.  
668     doi:10.3390/w12010169.

669     Vermote, E., Justice, C., Claverie, M., Franch, B., 2016. Preliminary analysis  
670     of the performance of the Landsat 8/OLI land surface reflectance product.  
671     *Remote Sensing of Environment* 185, 46–56. doi:10.1016/j.rse.2016.0  
672     4.008.

673     Williamson, C.E., Saros, J.E., Vincent, W.F., Smol, J.P., 2009. Lakes and  
674     reservoirs as sentinels, integrators, and regulators of climate change. *Lim-  
675     nology and Oceanography* 54, 2273–2282. doi:10.4319/lo.2009.54.6\_p  
676     art\_2.2273.

677 Zhang, Y., Shi, K., Sun, X., Zhang, Y., Li, N., Wang, W., Zhou, Y., Zhi,  
678 W., Liu, M., Li, Y., 2022. Improving remote sensing estimation of Secchi  
679 disk depth for global lakes and reservoirs using machine learning methods.  
680 GIScience & Remote Sensing 59, 1367–1383. doi:10.1080/15481603.202  
681 2.2116102.